Capstone checklist

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The brief: pick a task a computer can check → measure a small model on it → train it with GRPO → measure again on new puzzles → share the model and a short write-up.

1. Is my task a good one?

- ☐ **Checkable:** a short Python function can say right or wrong
- ☐ **Generatable:** a function can make thousands of puzzles with answers
- ☐ **Right difficulty:** the base model gets roughly 5–60% right (not 0%, not ~100%)
- ☐ **Short answers:** the final answer fits in a few tokens

2. Milestones

#	Milestone	Done when…
M0	Set up	`python selftest.py` passes on Colab (T4 GPU)
M1	Task + reward	Builder works; reward passes your tests, including cheats *and* correct answers written oddly
M2	Baseline	`evaluate.py` on 200 test puzzles: accuracy, format, length; 10 replies read
M3	Pilot	50 steps: `reward` moves, `frac_reward_zero_std` < 1.0, `completions/clipped_ratio` small
M4	Full run	300+ steps; reward and length curves saved
M5	Evaluate + share	Same test set before/after; 3 samples (1 wrong); model on Hugging Face

3. Commands

```
python selftest.py
python evaluate.py --task countdown --runs base=Qwen/Qwen2.5-0.5B-Instruct
python train.py --task countdown --max-steps 50          # pilot
python train.py --task countdown --max-steps 300         # full run
python evaluate.py --task countdown --runs base=Qwen/Qwen2.5-0.5B-Instruct
trained=outputs/countdown
```

4. What to submit

- **Hugging Face model** (a LoRA adapter is fine), with a model card: base model, task, rewards, settings, how to try it
- **Write-up, 300–800 words:**

Section	Include
Task	What puzzles look like; why they're checkable
Reward	What earns reward; one loophole you tested
Baseline	Accuracy, format, length; the command
Training	Steps, group size, learning rate, length cap; curves
Results	Before/after table (same test set); 3 real samples, 1 wrong
Reflection	A surprise, a limitation, a next experiment

5. Rubric at a glance (20 points: 14 pass, 18 distinction)

Task & reward · Baseline · Training run · Results, honestly · Reflection (4 points each)

Hard rules: test puzzles never in training · only numbers you measured (label others "illustrative") · model card names the base model.

6. If stuck

Symptom	Try
Reward flat at 0	Easier puzzles; read replies; test checker on correct answers
Reward up, accuracy flat	Loophole: read high-reward replies, tighten the checker
Out of memory	`--max-completion-length 192` or `--num-generations 4`
Replies keep growing	Lower the cap; add a gentle length reward
Late accuracy drop	Evaluate an earlier checkpoint (saved every 100 steps)